Initialisation

Start simulation with input:
- x_1 the activity in pixel 1
- x_2 the activity in pixel 2
- t the measurement time
- k the total number of simulation runs

Determine theoretical values for ML and LS (expected value, bias, variance, CR, MSE)

Simulation and computation

Generate 3 Poisson random variables (simulated data) with parameters:
- $y_1 = \text{Poisson}(x_1)$
- $y_2 = \text{Poisson}(x_2)$
- $y_3 = \text{Poisson}(x_1+x_2)$

Determine:
- ML estimators
- LS estimators

Repeat k times

Store value of:
- ML estimators
- LS estimators

After k times

Split the samples of estimators of size k in $\sqrt{k}$ subsamples of $\sqrt{k}$ estimators and determine:
- Expected value of the estimator in subsample
- Bias of the estimator in subsample

Results

Determine based on samples of estimators of size k:
- Observed exp. value of estimators
- Observed bias of estimators
- Observed variance of estimators
- Observed std. dev. of estimators
- Observed MSE of estimators

Determine based on samples of biasses of size $\sqrt{k}$:
- Expected value of biasses
- Standard deviation of biasses